

Healpy Exercise

1 Read a CMB map and its metadata

To read a CMB map in Healpy, the `healpy.read_map` function can be used. This function allows to load the map data from a FITS file and to mask the unwanted regions, plotted in Figure 1.

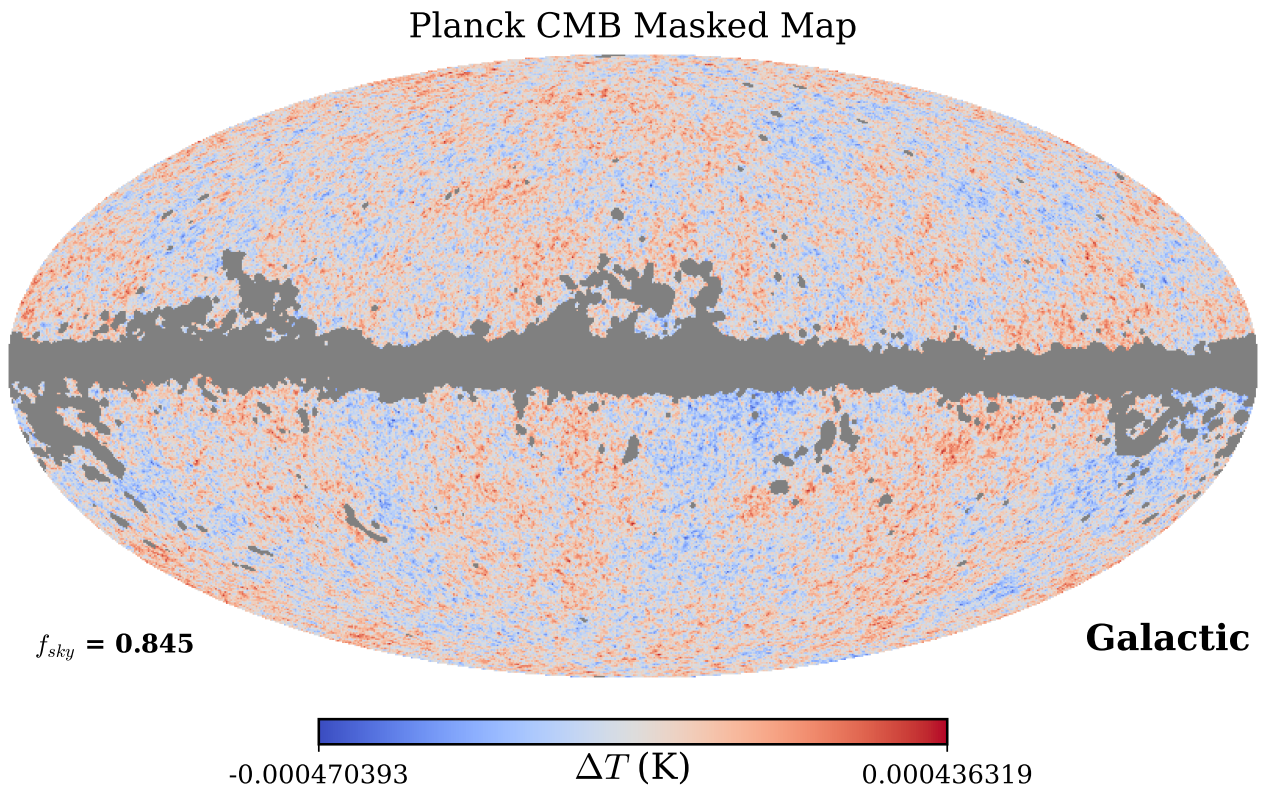


Figure 1: Mollview projection of the Planck CMB masked map in Galactic coordinates. The map has 2048 pixels along the side ($NSIDE = 2048$) and a resolution of 1.718 arcminutes. The color scale represents temperature fluctuations in kelvins (K).

The sky fraction that remains unmasked is called $f_{sky} = 0.845$. It is calculated as the mean of the squared mask values.

2 Apply some corrections to the raw data

Three different corrections are asked to be applied to the raw data in order to obtain a more accurate estimation of the CMB power spectrum (C_ℓ). The maximum multipole moment ℓ_{max} is set to 1500. The corrections are the following:

1. Correction for the effect of the beam is applied by removing the smoothing kernel:

$$C_\ell^{\text{smoothed}} = C_\ell^{\text{true}} \mathcal{B}^2(\ell), \quad (1)$$

where the beam window function $\mathcal{B}(\ell)$ is obtained from the following Healpy function: `healpy.sphtfunc.gauss_beam`.

2. Correction for the effect of the pixelisation is applied by removing the pixelisation window function:

$$C_\ell^{\text{pixelated}} = C_\ell^{\text{true}} \mathcal{C}^2(\ell), \quad (2)$$

where the pixelisation window function $\mathcal{C}(\ell)$ is obtained from the next Healpy function: `healpy.sphtfunc.pixwin`.

3. The effect of the mask is quite involved and induces a nontrivial modemixing, therefore it is forgone in first approximation. The C_ℓ will be rescaled like this:

$$C_\ell^{\text{masked}} = f_{\text{sky}} C_\ell^{\text{true}}. \quad (3)$$

Additionally, the multipoles $\ell < 30$ are discarded as they would have been most affected by the modemixing.

The data resulting of the last correction is binned in multipole space to reduce noise and improve visualization. The bin size is set to 30, and the binned power spectrum is calculated by averaging the C_ℓ values within each bin. The first bin starts at $\ell = 30$ to avoid the most affected multipoles by the mask, as mentioned before. Then, the binned power spectrum is rescaled to

$$D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell. \quad (4)$$

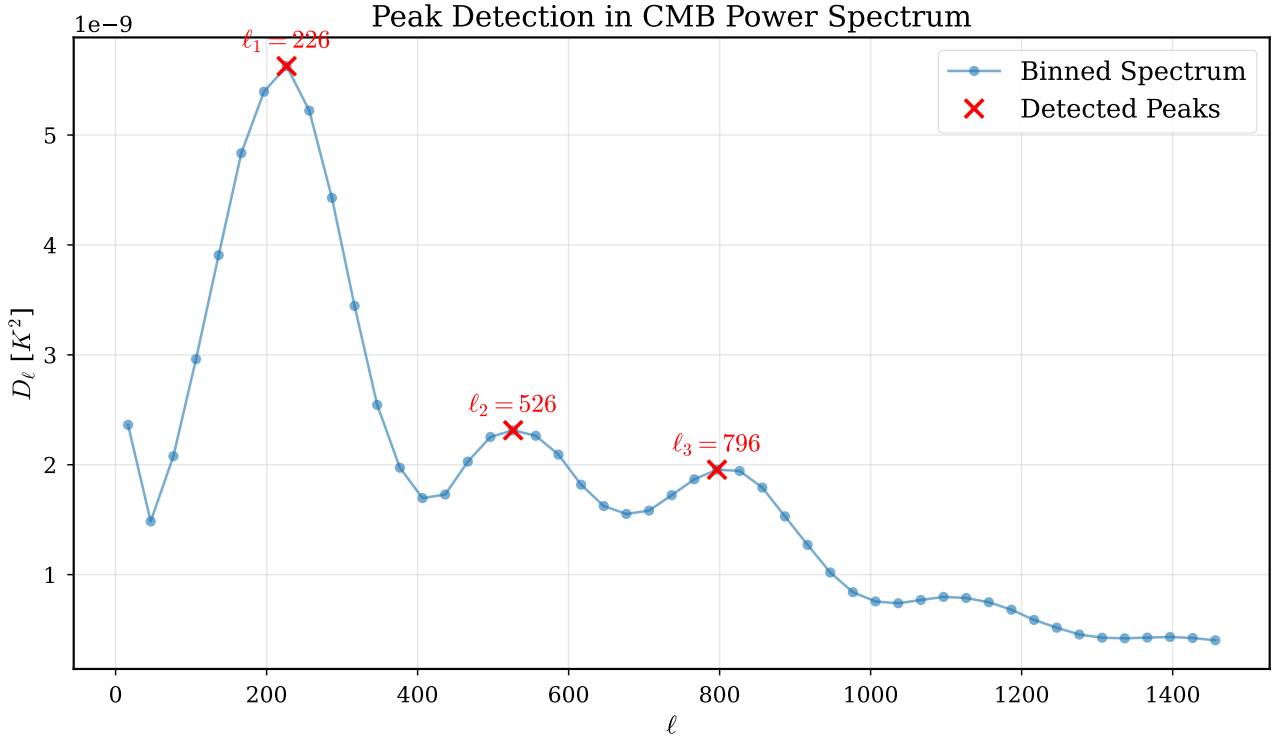


Figure 2: Binned CMB Power Spectrum after applying the corrections for mask. The power spectrum is represented in terms of Eq. (4). The bin size is 30 multipoles and the peaks are located at the multipole moments of $\ell = 226$, $\ell = 526$ and $\ell = 796$.

3 Analyse the data and extract physical information

Finally, the angular scale of the sound horizon during recombination, Θ_* , is estimated using two different methods based on the positions of the acoustic peaks in the CMB power spectrum.

1. Using relation of the first acoustic peak (first multipole, ℓ_1) with angular scales:

$$\Theta_* = \frac{\pi}{\ell_1}. \quad (5)$$

2. Using the spacing between the acoustic peaks:

$$\Delta\ell \approx \frac{\pi}{\Theta_*}. \quad (6)$$

The results from both methods are compared to assess their consistency in Table 1.

Method	Θ_* (rad)	Θ_* (deg)	Rel. Difference
First peak ($\ell_1 = 227$)	0.01387	0.79470	20.54%
Peak separation ($\Delta\ell = 285$)	0.01102	0.63158	

Table 1: Comparison of the angular scale of the sound horizon during recombination, Θ_* , estimated using two different methods based on the positions of the acoustic peaks in the CMB power spectrum.

The difference between the two estimates of Θ_* arises from the simplifications made in the second method. The first method directly relates the position of the first acoustic peak to the angular scale, while the second method assumes a constant spacing between peaks, which is just an approximation.